

Capstone Report: SecureWave Technology

Melvin Viado, Tim Loudon, and Daniel Lindo

Durham College

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Kristen MacDonald

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Introduction

SecureWave Technology utilizes and operates in a Microsoft domain environment supported by Windows Server 2019 that provides file sharing and network management capabilities. For context, a Microsoft Domain Environment provides a foundation for many businesses by using Active Directory Domain Services (ADDS) to provide a centralized management solution for a network of computers, users, and resources. In our scenario, SecureWave Technology has serious security challenges from irregular system updates, the absence of formal patch management policies, and outdated firmware on important network devices. These security challenges present risks including system instability, violations of compliance regulations, and increased vulnerability to cybercriminal exploitation. This report elaborates into these vulnerabilities comprehensively and provides actionable recommendations aimed at enhancing the security measures of SecureWave while ensuring uninterrupted operational continuity.

Security Vulnerabilities

Sporadic Patch Management

What are the potential risks if the vulnerability is not addressed?

SecureWave Technology's first vulnerability is that updates are being performed sporadically and inconsistently because there are no defined policies or procedures for updating their systems. The absence of routine updates on systems in a Microsoft domain environment exposes this organization to many severe cybersecurity threats, especially exploits of unpatched vulnerabilities. If SecureWave Technology does not implement a regular patch management system, the risk of attackers exploiting a vulnerability severely increases. Relating to this scenario, unpatched software is 5th in the top 10 threat vulnerabilities as seen in *Figure 1*.

Figure 1

Top 10 Threat Vulnerabilities



Note. From Top 10 Threat Vulnerabilities. (2025). DeepStrike. Retrieved 2025, (https://cdn.sanity.io/images/a3jopls3/testdataset/64db5c1b9a1a822d0b419f6e83c35b991c13d843-1536x1024.jpg).

There are also risks with the sporadic patch management itself as the patching process introduces security, stability, and operational issues due to undefined policies and procedures on system updates (*Patch Management Risks 2025*). Systems can experience instability, data corruption, or outages from complex patches with misunderstood prerequisites and invalid or malicious patches. Insufficient testing and poor planning on patches can break integrations, protocols, and cause excessive downtime and service interruptions. Lastly, SecureWave Technology does not mention the use of rollback procedures that can assist in reversing a faulty patch and waiting for a corrective update.

Common Vulnerabilities and Exposures (CVE) is a standardized dictionary of publicly known cybersecurity vulnerabilities in software and hardware. An example of a vulnerability is CVE-2024-49113, an Active Directory Lightweight Directory Access Protocol (LDAP) vulnerability that can crash unpatched Windows servers, enable remote code execution, and bypass traditional defense steps and directly target domain controllers (*How unpatched active directory flaws expose your Microsoft servers 2025*).

SecureWave Technology updates their servers by connecting directly to Microsoft's update servers. Since this organization does not use Windows Server Update Servers, which manages and deploys updates throughout a Microsoft domain environment, a man-in-the-middle attack can occur if certificate-pinned connections are interfered by Transport Layer Security proxies. This type of attack presents a high risk when proxy or firewall exceptions are misconfigured on Windows domain environment devices, which are used to create a valid connection to Microsoft Windows Update servers.

Finally, SecureWave Technology may experience compliance and operational impacts associated with this risk. Failure to effectively patch their systems violates regulations, increases risks fines and audits, and loses business trust. Depending on where SecureWave Technology conducts their business, some applicable regulations may involve the General Data Protection Regulation (GDPR) or the Personal Information Protection and Electronic Documents Act (PIPEDA). Operationally, this organization may not continue to effectively deliver their services to customers by not addressing this risk that leads to system instability, downtime from existing bugs, and compatibility issues.

How can the vulnerability be addressed?

Performing routine upgrades on all systems prevents vulnerabilities to commonly published exploits (MacDonald, *Security Vulnerabilities in a Windows Environment* 2025). These updates patch security flaws, bugs, and weaknesses in systems and applications before attackers can exploit them. These updates also improve compatibility, authentication protocols, and system integrity while reducing the attack surface of the environment. Patches should be applied along with the assessment of risks across the business lifecycle. Patches for known critical vulnerabilities should be applied as soon as possible.

Regularly scanning for vulnerabilities on systems using tools like Nmap helps with identifying outdated services and misconfigurations before attacks occur. Subscribing to security bulletins related to the technology solutions used by the organization can provide timely details on vulnerabilities and recommended patches. These bulletins also ensure compliance and provide guidance to hardware and software hardening, effectively covering security gaps in Windows environments. Routine vulnerability scanning and use of security bulletins helps SecureWave Technology follow a proactive security posture by minimizing unpatched attack surfaces.

Prioritization is important in this scenario, as SecureWave Technology must prioritize patching their Windows Servers and domain controllers first. Otherwise, compensating controls such as Active Directory Lightweight Directory Access Protocol firewalls and Remote Procedure Call firewalls can be implemented for servers that cannot be immediately patched (*How unpatched active directory flaws expose your Microsoft servers* 2025). As a last resort, monitoring exploit code activity from various internet sources can show signs of a new vulnerability given its relevance to Microsoft domain environments.

Effective risk management requires continuous analysis of various data sources to determine which vulnerabilities to address first, which may involve exploitation trends, public disclosures, and threat actor behavior (*Patch Management Risks* 2025). Advanced vulnerability remediation, such as Vicarius, offers a service that integrates threat intelligence, vulnerability assessment, and patch management as a centralized platform to assist cybersecurity and information technology operators. With the correct vulnerability information, risk-based prioritization, and planned remediation steps, SecureWave Technology will effectively address this vulnerability.

Neglecting Firmware Updates on Network Devices

What are the potential risks if the vulnerability is not addressed?

SecureWave's second vulnerability is that their outdated firmware for their routers and switches within their network infrastructure. Outdated firmware for routers and switches is incredibly dangerous, as if the manufacturer has included security patches and fixes in the new firmware that the routers and switches do not currently have, that makes them a critical target for hackers desiring to breach the network or even reroute portions of internal traffic outside the network. This is especially dangerous if the routers and switches are managed, as this leaves a vulnerability for the attackers to exploit, by inputting new commands that can entirely change the behavior of packet delivery within the network. According to (Dirk-NetworkGuides, 2025 Key Takeaways), up to 70% of router vulnerabilities are due to outdated firmware.

Depending on the age of the networking devices along with their brand, networking protocols can be at risk and multiple devices on the network would also be at risk of tampering or having their traffic re-routed. Network critical devices or security implementations can become exploited or made useless if access to the routers and switches are gained through outdated firmware vulnerabilities.

Another risk is that the outdated firmware can result in poor connection performance for the network. The Firmware is responsible for optimizing the network usage, ensuring that devices have efficient and stable connection speeds (Acrylic, 2025 Risk of Obsolete Firmware). If the firmware is outdated, this can cause a bottleneck within the network, inhibiting the router from providing efficient data speeds and even stable connections to modern end devices. Outdated firmware can also result in incompatibility with new devices, as the protocols being used on the aged hardware may be so old, that that the modern device will not support it due to security concerns, or a simple situation of the modern device not supporting any old protocols in effect on the outdated hardware and firmware.

Figure 2

A Device Portraying Absence of Connection in the Network



Outdated firmware can also lead to network outages, especially if modern devices with new firmware are being utilized among the outdated devices. This can also lead to security solutions for the network malfunctioning, such as a SIEM (Security Information and Event Manager). A SIEM is responsible for monitoring network traffic among routers, switches firewalls, end devices and servers on the network. Routers and switches within the network pose a security risk due to security standards and protocols that may not be supported on new devices. This also leaves a door open for attackers as any known vulnerability found in the networking devices with old firmware will be exploited, leaving an area of the network unmonitored which the SIEM cannot possibly compensate for, such as outdated protocols in use. The outdated devices can also increase the risk of false positives or have the SIEM turn a blind eye to suspicious activity due to old firmware standards.

How can the vulnerability be addressed?

When it comes to outdated firmware on networking devices, not many options are available. Though the options are few, what can be implemented is powerful. Firstly, SecureWave Technology can start by making routine upgrades to the routers and switches in the network to counter the threat of vulnerabilities and exploits that are found in the firmware of old hardware. Old versions of hardware should be routinely tested for vulnerabilities, and if found to be vulnerable, the device should be recycled or destroyed and replaced as soon as possible.

The second option that can be taken would be to simply update the outdated firmware on the networking devices, to ensure that critical patches are applied efficiently and in a risk-based fashion. Regular inventory of the networking devices should also be maintained along with the relevant version number of the device. Consistent housekeeping of the network infrastructure in relation to routers and switches should be made regularly, as these devices are the backbone of the network, and their compromise can quickly lead to the breached security of other devices on the network. If the routers and switches are insecure, nothing on the network is safe from the attackers reach unless it is highly secured or air gapped from the network.

Figure 3

An Example of a Network Device (Router) that Requires Updates on Firmware



Lack of Change Management and Documentation

What are the potential risks if the vulnerability is not addressed?

When working in an enterprise environment like SecureWave has, a critical way to make sure that updating your systems is done securely is through having a proxy server that the company's machines (both physical and virtual) can connect to for updates rather than connecting directly to the service provider for updates. This is done because proxy servers such as ones configured to run Windows Server Update Services (WSUS) allow administrators to roll out updates with a more granular level of control as to what devices will receive which updates and when, following an implemented update policy.

Updating network devices is one of the best ways to keep them secure against attackers looking to breach systems and tamper with sensitive data. SecureWave's practices of only updating their systems sporadically and using various versions of security patches is very unsecure and opens them up to various cyberattacks such as zero-day vulnerabilities and other problems such as compatibility issues.

According to Xantaro, not updating software across all devices on a regular basis can open them up to zero-day attacks because "After the release of an update, such vulnerabilities are known and can be actively exploited by attackers. It is therefore crucial to install updates as soon as they are released." By keeping devices secure through updating software and firmware, they become more resistant to a wide variety of attacks and exploits and makes known threats less of a risk against systems like those of SecureWave's. Since SecureWave is downloading their critical security patches by connecting directly to Microsoft's servers, they have the potential of opening themselves up to other attacks such as man-in-the-middle exploits if the server they are connecting to is not truly a Microsoft update server.

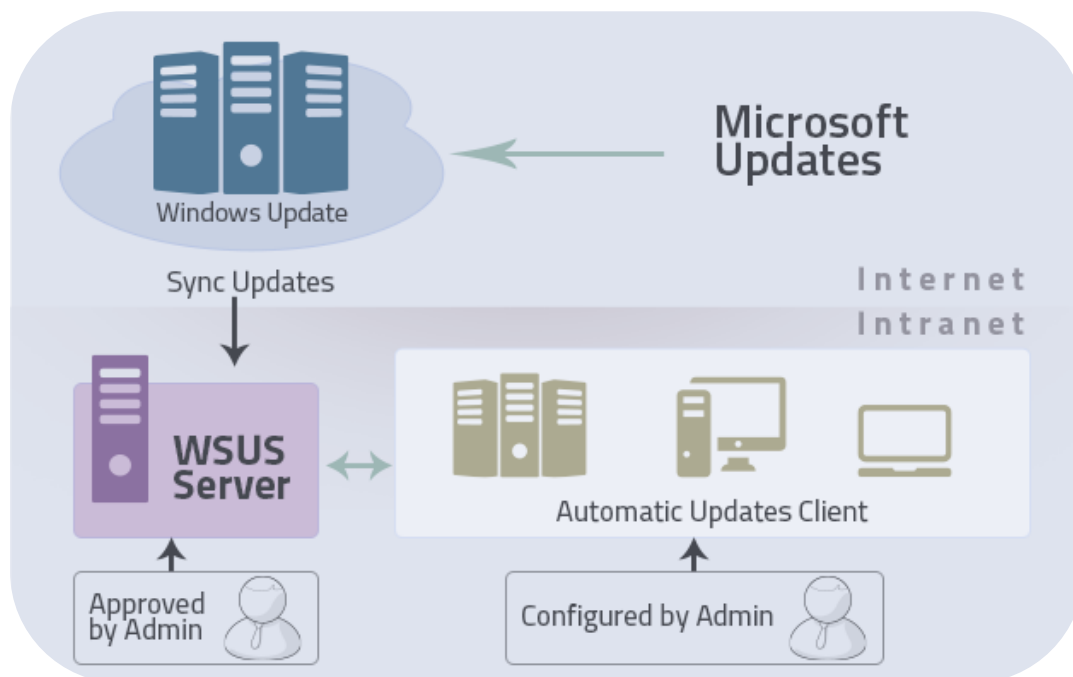
How can the vulnerability be addressed?

Because SecureWave is using Windows machines and Windows Server to run its systems, they can take advantage of Windows Server Update Services (WSUS) and use a proxy server configured with it to deploy updates to their devices simultaneously. This is a more secure way to update devices and software as opposed to what SecureWave is currently doing because “Update payloads are protected against modification by multiple means, including digital signature checks and cryptographic hash verifications” (*Ten Forums, Improving security for Windows 10 devices receiving updates via WSUS*) using tools such as HTTPS.

By automating these updates through WSUS, you can centralize and automate update management for Microsoft products. This helps you determine how and when to distribute updates and which machines require a specific update (*Solar Winds, What is WSUS? windows server update services guide*). This provides more control as to which devices get certain updates and will ensure that all network devices are kept up-to-date, secure and maintain compatibility with each other.

Figure 4

Flow Chart of Patch Management using Windows Server Update Services



Conclusion

SecureWave Technology's current approach to their systems and network maintenance shows they have a vulnerable environment with severe cybersecurity threats. The absence and lack of structured update policies, reliance on direct connections to Microsoft update servers, and outdated firmware on routers and switches create exploitable vulnerabilities that could compromise sensitive data and business operations. By implementing patch management cycle solutions, enforcing documented change control processes, and maintaining updated firmware across all network devices, SecureWave can greatly reduce its attack surface. Additionally, proactive security measures and continuous monitoring will enable the organization to safeguard its infrastructure and maintain trust with clients and stakeholders in a high-risk digital landscape.

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